

Activity_Discover what is in your dataset

April 17, 2025

1 Activity: Discover what is in your dataset

1.1 Introduction

In this activity, you will discover characteristics of a dataset and use visualizations to analyze the data. This will develop and strengthen your skills in **exploratory data analysis (EDA)** and your knowledge of functions that allow you to explore and visualize data.

EDA is an essential process in a data science workflow. As a data professional, you will need to conduct this process to better understand the data at hand and determine how it can be used to solve the problem you want to address. This activity will give you an opportunity to practice that process and prepare you for EDA in future projects.

In this activity, you are a member of an analytics team that provides insights to an investing firm. To help them decide which companies to invest in next, the firm wants insights into **unicorn companies**—companies that are valued at over one billion dollars. The data you will use for this task provides information on over 1,000 unicorn companies, including their industry, country, year founded, and select investors. You will use this information to gain insights into how and when companies reach this prestigious milestone and to make recommendations for next steps to the investing firm.

1.2 Step 1: Imports

1.2.1 Import libraries and packages

First, import relevant Python libraries and modules. Use the `pandas` library and the `matplotlib.pyplot` module.

```
[1]: # Import libraries and packages

import pandas as pd
import matplotlib.pyplot as plt
```

1.2.2 Load the dataset into a DataFrame

The dataset provided is in the form of a csv file named `Unicorn_Companies.csv` and contains a subset of data on unicorn companies. As shown in this cell, the dataset has been automatically

loaded in for you. You do not need to download the .csv file, or provide more code, in order to access the dataset and proceed with this lab. Please continue with this activity by completing the following instructions.

```
[2]: # RUN THIS CELL TO IMPORT YOUR DATA.

companies = pd.read_csv("Unicorn_Companies.csv")
```

Hint 1

Refer to what you learned about [loading data](#) in Python.

Hint 2

Use the function in the **pandas** library that allows you to read data from a csv file and load the data into a DataFrame.

Hint 3

Use the `read_csv()` function from the **pandas** library.

1.3 Step 2: Data exploration

1.3.1 Display the first 10 rows of the data

Next, explore the dataset and answer questions to guide your exploration and analysis of the data. To begin, display the first 10 rows of the data to get an understanding of how the dataset is structured.

```
[3]: # Display the first 10 rows of the data

companies.head(10)
```

```
[3]:
```

	Company	Valuation	Date Joined	Industry \
0	Bytedance	\$180B	4/7/17	Artificial intelligence
1	SpaceX	\$100B	12/1/12	Other
2	SHEIN	\$100B	7/3/18	E-commerce & direct-to-consumer
3	Stripe	\$95B	1/23/14	Fintech
4	Klarna	\$46B	12/12/11	Fintech
5	Canva	\$40B	1/8/18	Internet software & services
6	Checkout.com	\$40B	5/2/19	Fintech
7	Instacart	\$39B	12/30/14	Supply chain, logistics, & delivery
8	JUUL Labs	\$38B	12/20/17	Consumer & retail
9	Databricks	\$38B	2/5/19	Data management & analytics

	City	Country/Region	Continent	Year Founded	Funding \
0	Beijing	China	Asia	2012	\$8B
1	Hawthorne	United States	North America	2002	\$7B
2	Shenzhen	China	Asia	2008	\$2B

3	San Francisco	United States	North America	2010	\$2B
4	Stockholm	Sweden	Europe	2005	\$4B
5	Surry Hills	Australia	Oceania	2012	\$572M
6	London	United Kingdom	Europe	2012	\$2B
7	San Francisco	United States	North America	2012	\$3B
8	San Francisco	United States	North America	2015	\$14B
9	San Francisco	United States	North America	2013	\$3B

Select Investors

0	Sequoia Capital China, SIG Asia Investments, S...
1	Founders Fund, Draper Fisher Jurvetson, Rothen...
2	Tiger Global Management, Sequoia Capital China...
3	Khosla Ventures, LowercaseCapital, capitalG
4	Institutional Venture Partners, Sequoia Capita...
5	Sequoia Capital China, Blackbird Ventures, Mat...
6	Tiger Global Management, Insight Partners, DST...
7	Khosla Ventures, Kleiner Perkins Caufield & By...
8	Tiger Global Management
9	Andreessen Horowitz, New Enterprise Associates...

Hint 1

Refer to [the content about exploratory data analysis in Python](#).

Hint 2

Use the function in the `pandas` library that allows you to get a specific number of rows from the top of a `DataFrame`.

Hint 3

Use the `head()` function from the `pandas` library.

Question: What do you think the “Date Joined” column represents?

The “Date Joined” column represents the date that the company became a unicorn company.

Question: What do you think the “Select Investors” column represents?

The “select investor” typically refers to investors that are part of a private or restricted investment group. In this case it represents the top investors in the company.

1.3.2 Assess the size of the dataset

Get a sense of how large the dataset is. The `size` property that `DataFrames` have can help.

```
[4]: # How large the dataset is

companies.size
```

```
[4]: 10740
```

Question: What do you notice about the size of the dataset?

The dataset contains 10740 entries.

1.3.3 Determine the shape of the dataset

Identify the number of rows and columns in the dataset. The `shape` property that DataFrames have can help.

```
[5]: # Shape of the dataset
companies.shape
```

```
[5]: (1074, 10)
```

Question: What do you notice about the shape of the dataset?

The shape function identifies the number of rows (1074) and the number of columns (10).

1.3.4 Get basic information about the dataset

To further understand what the dataset entails, get basic information about the dataset, including the data type of values in each column. There is more than one way to approach this task. In this instance, use the `info()` function from `pandas`.

```
[11]: # Get information
companies.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1074 entries, 0 to 1073
Data columns (total 10 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Company               1074 non-null   object
 1   Valuation             1074 non-null   object
 2   Date Joined           1074 non-null   object
 3   Industry              1074 non-null   object
 4   City                  1058 non-null   object
 5   Country/Region        1074 non-null   object
 6   Continent             1074 non-null   object
 7   Year Founded          1074 non-null   int64
 8   Funding               1074 non-null   object
 9   Select Investors      1073 non-null   object
dtypes: int64(1), object(9)
memory usage: 84.0+ KB
```

Question: What do you notice about the type of data in the `Year Founded` column? Refer to the output from using `info()` above. Knowing the data type of this variable is helpful because

it indicates what types of analysis can be done with that variable, how it can be aggregated with other variables, and so on.

The “year found” column contains int64 values. This means that the column represents an integer.

Question: What do you notice about the type of data in the `Date Joined` column? Refer to the output from using `info()` above. Knowing the data type of this variable is helpful because it indicates what types of analysis can be done with that variable and how the variable can be transformed to suit specific tasks.

The “Date Joined” column contains objects. This means that the date is represented as an object.

1.4 Step 3: Statistical tests

1.4.1 Find descriptive statistics

Find descriptive statistics and structure your dataset. The `describe()` function from the `pandas` library can help. This function generates statistics for the numeric columns in a dataset.

```
[12]: # Get descriptive statistics
      companies.describe()
```

```
[12]:      Year Founded
count    1074.000000
mean     2012.895717
std        5.698573
min      1919.000000
25%      2011.000000
50%      2014.000000
75%      2016.000000
max      2021.000000
```

Question: Based on the table of descriptive stats generated above, what do you notice about the minimum value in the `Year Founded` column? This is important to know because it helps you understand how early the entries in the data begin.

The minimum value in the data set is 1919 meaning that the first company recorded in the data set became a unicorn company in the year 1919. Meaning that the data set contains 106 years of data.

Question: What do you notice about the maximum value in the `Year Founded` column? This is important to know because it helps you understand the most recent year captured by the data.

The maximum value show us that the last company to became a unicorn, was in 2021.

1.4.2 Convert the `Date Joined` column to datetime

Use the `to_datetime()` function from the `pandas` library to convert the `Date Joined` column to datetime. This splits each value into year, month, and date components. This is an important step

in data cleaning, as it makes the data in this column easier to use in tasks you may encounter. To name a few examples, you may need to compare “date joined” between companies or determine how long it took a company to become a unicorn. Having “date joined” in datetime form would help you complete such tasks.

```
[6]: # Step 1: Use pd.to_datetime() to convert Date Joined column to datetime
      # Step 2: Update the column with the converted values
```

```
companies['Date Joined'] = pd.to_datetime(companies['Date Joined'])
```

```
[7]: # Use .info() to confirm that the update actually took place
```

```
companies.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1074 entries, 0 to 1073
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Company                1074 non-null   object
1   Valuation              1074 non-null   object
2   Date Joined            1074 non-null   datetime64[ns]
3   Industry               1074 non-null   object
4   City                   1058 non-null   object
5   Country/Region         1074 non-null   object
6   Continent              1074 non-null   object
7   Year Founded           1074 non-null   int64
8   Funding                1074 non-null   object
9   Select Investors       1073 non-null   object
dtypes: datetime64[ns](1), int64(1), object(8)
memory usage: 84.0+ KB
```

1.4.3 Create a Year Joined column

It is common to encounter situations where you will need to compare the year joined with the year founded. The `Date Joined` column does not just have year—it has the year, month, and date. Extract the year component from the `Date Joined` column and add those year components into a new column to keep track of each company’s year joined.

```
[8]: # Step 1: Use .dt.year to extract year component from Date Joined column
      # Step 2: Add the result as a new column named Year Joined to the DataFrame
```

```
companies['Year Joined'] = companies['Date Joined'].dt.year
```

```
[18]: # Use .head() to confirm that the new column did get added
```

```
companies.head(10)
```

```
[18]:      Company Valuation Date Joined      Industry \
0      Bytedance      $180B 2017-04-07      Artificial intelligence
1      SpaceX      $100B 2012-12-01      Other
2      SHEIN      $100B 2018-07-03      E-commerce & direct-to-consumer
3      Stripe      $95B 2014-01-23      Fintech
4      Klarna      $46B 2011-12-12      Fintech
5      Canva      $40B 2018-01-08      Internet software & services
6      Checkout.com      $40B 2019-05-02      Fintech
7      Instacart      $39B 2014-12-30      Supply chain, logistics, & delivery
8      JUUL Labs      $38B 2017-12-20      Consumer & retail
9      Databricks      $38B 2019-02-05      Data management & analytics
```

```
      City Country/Region      Continent Year Founded Funding \
0      Beijing      China      Asia      2012      $8B
1      Hawthorne      United States      North America      2002      $7B
2      Shenzhen      China      Asia      2008      $2B
3      San Francisco      United States      North America      2010      $2B
4      Stockholm      Sweden      Europe      2005      $4B
5      Surry Hills      Australia      Oceania      2012      $572M
6      London      United Kingdom      Europe      2012      $2B
7      San Francisco      United States      North America      2012      $3B
8      San Francisco      United States      North America      2015      $14B
9      San Francisco      United States      North America      2013      $3B
```

```
      Select Investors      Year Joined
0      Sequoia Capital China, SIG Asia Investments, S...      2017
1      Founders Fund, Draper Fisher Jurvetson, Rothen...      2012
2      Tiger Global Management, Sequoia Capital China...      2018
3      Khosla Ventures, LowercaseCapital, capitalG      2014
4      Institutional Venture Partners, Sequoia Capita...      2011
5      Sequoia Capital China, Blackbird Ventures, Mat...      2018
6      Tiger Global Management, Insight Partners, DST...      2019
7      Khosla Ventures, Kleiner Perkins Caufield & By...      2014
8      Tiger Global Management      2017
9      Andreessen Horowitz, New Enterprise Associates...      2019
```

1.5 Step 4: Results and evaluation

1.5.1 Take a sample of the data

It is not necessary to take a sample of the data in order to conduct the visualizations and EDA that follow. But you may encounter scenarios in the future where you will need to take a sample of the data due to time and resource limitations. For the purpose of developing your skills around sampling, take a sample of the data and work with that sample for the next steps of analysis you want to conduct. Use the `sample()` function for this task.

- Use `sample()` with the `n` parameter set to 50 to randomly sample 50 unicorn companies from

the data. Be sure to specify the `random_state` parameter to ensure reproducibility of your work. Save the result to a variable called `companies_sampled`.

```
[9]: # Sample the data

companies_sample = companies.sample(n = 50 ,random_state = 42)
```

1.5.2 Visualize the time it took companies to reach unicorn status

Visualize the longest time it took companies to reach unicorn status for each industry represented in the sample. To create a bar plot to visualize this, use the `bar()` function from the `matplotlib.pyplot` module. You'll first need to prepare the data.

```
[10]: # Prepare data for plotting

# Create new 'years_till_unicorn' column

companies_sample['years_till_unicorn'] = companies_sample['Year Joined'] -
↳ companies_sample['Year Founded']

# Group the data by 'Industry'. For each industry get the max value for year in
↳ 'years_till_unicorn' column.

grouped = (companies_sample[['Industry', 'years_till_unicorn']]
           .groupby('Industry')
           .max()
           .sort_values(by='years_till_unicorn')
           )

grouped
```

```
[10]:
```

	years_till_unicorn
Industry	
Consumer & retail	1
Auto & transportation	2
Artificial intelligence	5
Data management & analytics	8
Mobile & telecommunications	9
Supply chain, logistics, & delivery	12
Internet software & services	13
Other	15
E-commerce & direct-to-consumer	18
Cybersecurity	19
Fintech	21
Health	21


```

[11]: # Create bar plot
      # with the various industries as the categories of the bars
      # and the time it took to reach unicorn status as the height of the bars

      plt.bar(grouped.index, grouped['years_till_unicorn'])

      # Set title

      plt.title("Bar Plot of maximum years taken by a company to become a unicorn per_
      ↳industry (from sample)")

      # Set x-axis label

      plt.xlabel('Industry')

      # Set y-axis label

      plt.ylabel('Max number of years')

      # Rotate labels on the x-axis as a way to avoid overlap in the positions of the_
      ↳text

      plt.xticks(rotation=45,horizontalalignment='right')

      # Display the plot

      plt.show

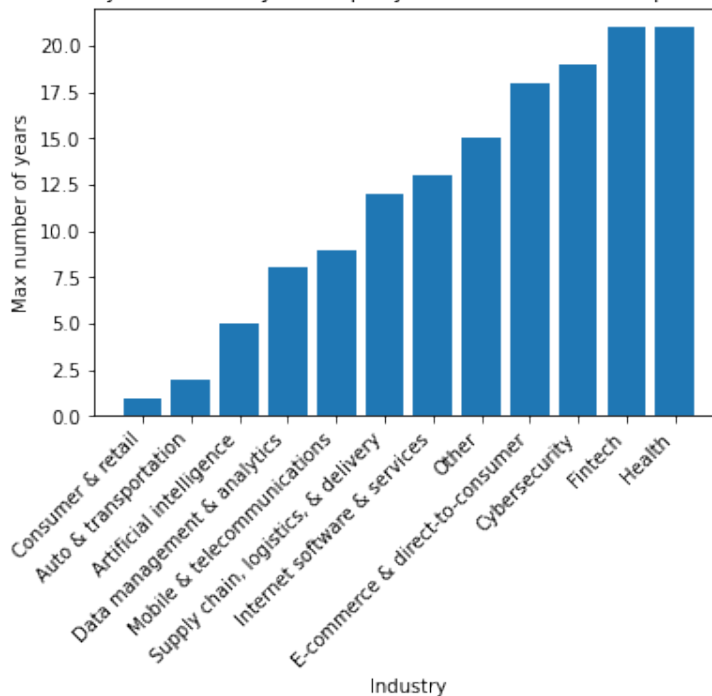
```

```

[11]: <function matplotlib.pyplot.show(*args, **kw)>

```

Bar Plot of maximum years taken by a company to become a unicorn per industry (from sample)



Hint 1

To prepare the data for modeling, begin by creating a column that represents the number of years it took each company to reach unicorn status.

Hint 2

To prepare the data for modeling, group the dataframe by industry and get the maximum value in the newly created column for each industry.

Hint 3

You can use the `plt.bar()` to create the bar plot, passing in the categories and heights of the bars.

You can use `plt.title()`, `plt.xlabel()`, and `plt.ylabel()` to set the title, x-axis label, and y-axis label, respectively.

You can use `plt.xticks()` to rotate labels on the x-axis of a plot. The parameters `rotation=45`, `horizontalalignment='right'` can be passed in to rotate the labels by 45 degrees and align the labels to the right.

You can use `plt.show()` to display a plot.

Question: What do you observe from this bar plot?

Response:

The bar graph shows the minimum and maximum amount of time that it took companies to become a unicorn. Companies in the fintech and healthcare industries take the longest amount of time

to become an unicorn and companies in consumer & retail industries take the shortest amount of time.

1.5.3 Visualize the maximum unicorn company valuation per industry

Visualize unicorn companies' maximum valuation for each industry represented in the sample. To create a bar plot to visualize this, use the `bar()` function from the `matplotlib.pyplot` module. Before plotting, create a new column that represents the companies' valuations as numbers (instead of strings, as they're currently represented). Then, use this new column to plot your data.

```
[13]: # Create a column representing company valuation as numeric data

# Create new column
companies_sample['valuation_billions'] = companies_sample['Valuation']
# Remove the '$' from each value
companies_sample['valuation_billions'] = companies_sample['valuation_billions'].
    ↪str.replace('$', '')
# Remove the 'B' from each value
companies_sample['valuation_billions'] = companies_sample['valuation_billions'].
    ↪str.replace('B', '')
# Convert column to type int
companies_sample['valuation_billions'] = companies_sample['valuation_billions'].
    ↪astype('int')
companies_sample.head()
```

```
[13]:
```

	Company	Valuation	Date Joined	Industry \
542	Aiven	\$2B	2021-10-18	Internet software & services
370	Jusfoun Big Data	\$2B	2018-07-09	Data management & analytics
307	Innovaccer	\$3B	2021-02-19	Health
493	Algolia	\$2B	2021-07-28	Internet software & services
350	SouChe Holdings	\$3B	2017-11-01	E-commerce & direct-to-consumer

	City	Country/Region	Continent	Year Founded	Funding \
542	Helsinki	Finland	Europe	2016	\$210M
370	Beijing	China	Asia	2010	\$137M
307	San Francisco	United States	North America	2014	\$379M
493	San Francisco	United States	North America	2012	\$334M
350	Hangzhou	China	Asia	2012	\$1B

	Select Investors	Year Joined \
542	Institutional Venture Partners, Atomico, Early...	2021
370	Boxin Capital, DT Capital Partners, IDG Capital	2018
307	M12, WestBridge Capital, Lightspeed Venture Pa...	2021
493	Accel, Alven Capital, Storm Ventures	2021
350	Morningside Ventures, Warburg Pincus, CreditEa...	2017

```
years_till_unicorn  valuation_billions
```

542	5	2
370	8	2
307	7	3
493	9	2
350	5	3

[16]: *# Prepare the data for modeling.*

```
grouped = (companies_sample[['Industry', 'valuation_billions']]
            .groupby('Industry')
            .max()
            .sort_values(by='valuation_billions')
            )
grouped
```

```
[16]:
```

	valuation_billions
Industry	
Auto & transportation	1
Consumer & retail	1
Other	2
Supply chain, logistics, & delivery	2
Cybersecurity	3
Health	3
Data management & analytics	4
E-commerce & direct-to-consumer	4
Internet software & services	5
Mobile & telecommunications	7
Fintech	10
Artificial intelligence	12

[19]: *# Create bar plot*
with the various industries as the categories of the bars
and the maximum valuation for each industry as the height of the bars

```
plt.bar(grouped.index, grouped['valuation_billions'])

# Set title

plt.title("Bar plot of maximum unicorn company valuation per industry (from_
→sample)")

# Set x-axis label

plt.xlabel('Industry')
```

```

# Set y-axis label

plt.ylabel('Maximum valuation in billions of dollars')

# Rotate labels on the x-axis as a way to avoid overlap in the positions of the
→ text

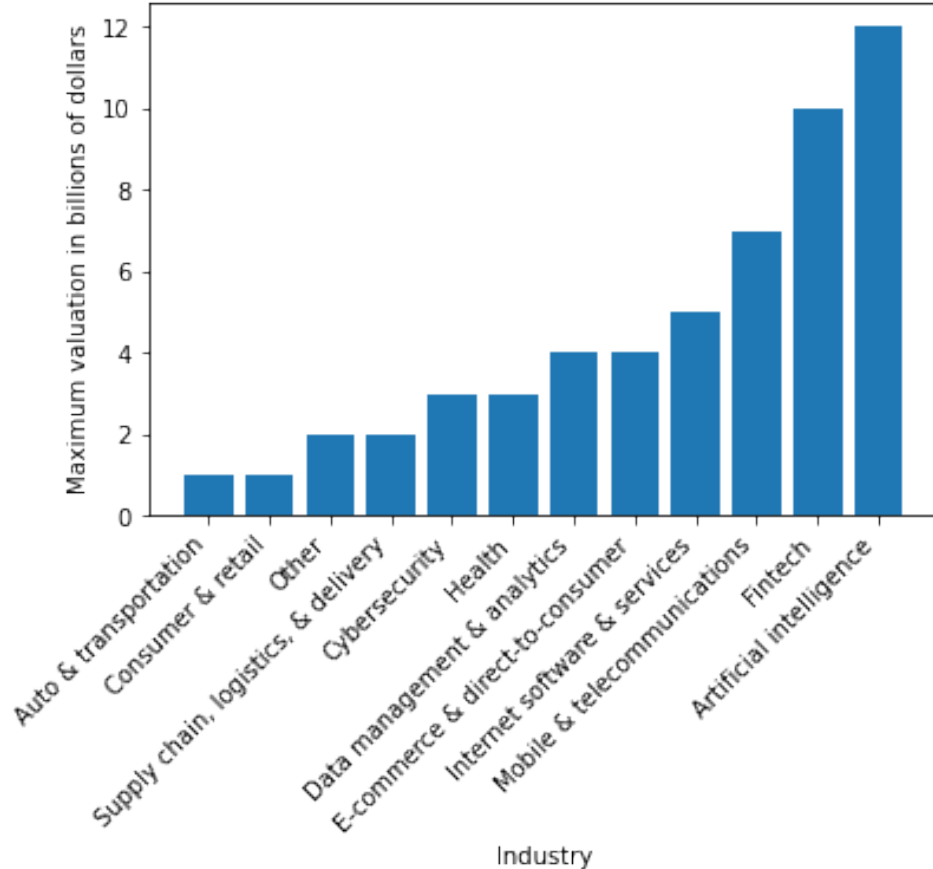
plt.xticks(rotation=45, horizontalalignment = 'right')

# Display the plot

plt.show()

```

Bar plot of maximum unicorn company valuation per industry (from sample)



Hint 1

Repeat the process from the last task, only this time with different variables.

Hint 2

Use the function in the `matplotlib.pyplot` module that allows you to create a bar plot, specifying the category and height for each bar.

Use the functions in the `matplotlib.pyplot` module that allow you to set the title, x-axis label, and y-axis label of plots. In that module, there are also functions for rotating the labels on the x-axis and displaying the plot.

Hint 3

Use the `plt.bar()` to create the bar plot, passing in the categories and heights of the bars.

Use `plt.title()`, `plt.xlabel()`, and `plt.ylabel()` to set the title, x-axis label, and y-axis label, respectively.

Use `plt.xticks()` to rotate labels on the x-axis of a plot. The parameters `rotation=45`, `horizontalalignment='right'` can be passed in to rotate the labels by 45 degrees and align the labels to the right.

Use `plt.show()` to display a plot.

Question: What do you observe from this bar plot?

This bar plot shows that for this sample, the highest maximum valuation occurred in the artificial intelligence industry, while the lowest valuation occurred in the Auto & transportation industry.

1.6 Considerations

What are some key takeaways that you learned from this lab?

- Functions in the pandas library can be used to gather the characteristics of the data at hand. The `info()` and `describe()` functions are especially useful for gathering basic information about the dataset and finding descriptive statistics.
- Functions in the `matplotlib.pyplot` module can be used to create bar charts and graphs to further understand aspects of the data. The `bar()` functions allow you to create bar graphs and help you visualise categorical information about the data. I was able to visualize the maximum number of years it took to become a unicorn for each industry in the sample representing the data.

What findings would you share with others?

- There were 1074 represented in this dataset.
- Some companies take longer than others to require unicorn status but have reached high valuation as of March 2022. Some companies may take longer to reach unicorn status for many reasons, including requiring more funding or taking longer to develop their business model.

What recommendations would you share with stakeholders based on these findings?

Focusing specifically on companies in a certain industry can be the next step:

- Identify the main industries the investment firm is interested in.

- Select a subset of data that include only companies in those industries.
- Analyse the subset closely. Determine which companies have higher value but do not have many investors. These companies might be good candidates to consider investing in.

References

Bhat, M.A. (2022, March). *Unicorn Companies*.

Congratulations! You've completed this lab. However, you may not notice a green check mark next to this item on Coursera's platform. Please continue your progress regardless of the check mark. Just click on the "save" icon at the top of this notebook to ensure your work has been logged.